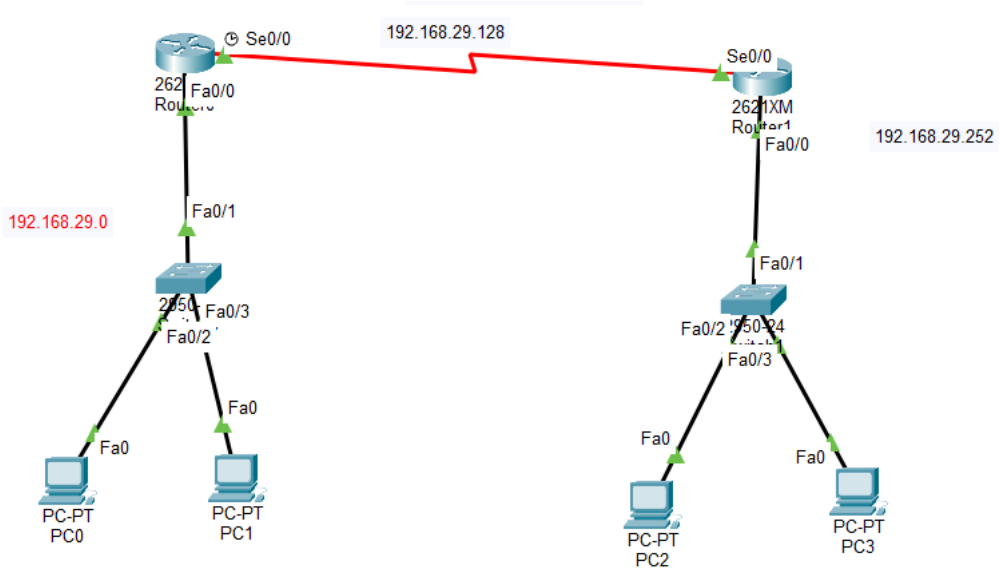
4-5: Weekly Practice and TroubleshootingPractical (2 hours):Resolve connectivity issues in a complex network.Task: Submit troubleshooting steps and resolutions.

Task 1.

Use the ping command to test connectivity between two PCs on different subnets in your Packet Tracer network and document the output for both successful and failed attempts.




```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.29.30

Pinging 192.168.29.30 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.29.30:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.29.190

Pinging 192.168.29.190 with 32 bytes of data:

Request timed out.
Reply from 192.168.29.190: bytes=32 time=5ms TTL=126
Reply from 192.168.29.190: bytes=32 time=30ms TTL=126
Reply from 192.168.29.190: bytes=32 time=6ms TTL=126

Ping statistics for 192.168.29.190:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 5ms, Maximum = 30ms, Average = 13ms

C:\>ping 192.168.29.190

Pinging 192.168.29.190 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.29.190:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Task 2.

Identify and fix an incorrect default gateway configuration on one device in your network so that it can access an external web server simulation; submit the steps you took and the command outputs before and after the fix.

Fix an incorrect default gateway

Objective

Identify an incorrect default gateway on a PC and restore access to an external web server simulation.

Step 1: Check the incorrect configuration

On PC0, open Desktop → Command Prompt:

ipconfig

Example output:

IPv4 Address: 192.168.10.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.10.254

Test the gateway:

ping 192.168.10.254

If the address is not assigned to a reachable device, the ping will fail. However, the failed ping alone does not prove that the gateway is incorrect.

Step 2: Correct the gateway

1. Click PC0.
2. Open Desktop → IP Configuration.
3. Change Default Gateway from 192.168.10.254 to 192.168.10.1.
4. Keep the PC's IP address and subnet mask unchanged.

Step 3: Verify after the fix

ipconfig

ping 192.168.10.1

ping 203.0.113.10

Expected results, assuming the router and server are configured correctly:

Default Gateway: 192.168.10.1

Reply from 192.168.10.1

Reply from 203.0.113.10

Then test the web service using the PC's Web Browser and enter the server's IP address.

For a working end-to-end connection, R1 also needs a valid route to the server network, and the return path must be configured. If using a simulated Internet connection, configure the server and router accordingly.

Lab report conclusion: The incorrect default gateway prevented the PC from forwarding traffic to remote networks. Correcting the gateway restored connectivity, provided the routing and web service were otherwise functional.

Task 3.

A user reports they cannot access Flipkart.com from their simulated network. Trace the path using traceroute (or tracert) and identify where the connection fails; explain how you diagnosed and resolved the issue.

```
C:\Users\sketan>nslookup flipkart.com
Server:  UnKnown
Address:  10.131.7.129

Non-authoritative answer:
Name:    flipkart.com
Address:  163.53.77.62


C:\Users\sketan>tracert flipkart.com

Tracing route to flipkart.com [163.53.77.62]
over a maximum of 30 hops:

  1     3 ms     1 ms     1 ms  10.131.7.129
  2     *        *        *     Request timed out.
  3    67 ms    38 ms    40 ms  10.0.242.113
  4    62 ms    55 ms    39 ms  125.17.187.161
  5   129 ms    61 ms    57 ms  116.119.33.80
  6    84 ms    69 ms    57 ms  125.19.147.10
  7     *        *        *     Request timed out.
  8     *        *        *     Request timed out.
  9     *        *        *     Request timed out.
 10     *        *        *     Request timed out.
 11    60 ms    58 ms    61 ms  163.53.77.62

Trace complete.
```

Step 1: Check DNS resolution

On a Packet Tracer PC, open Command Prompt:

nslookup flipkart.com

If name resolution fails, investigate DNS server configuration. If your Packet Tracer version does not support nslookup, verify DNS through the PC's IP Configuration and the simulated DNS server.

Step 2: Trace the route

On a Windows PC:

tracert flipkart.com

Alternatively, trace the server's IP directly:

tracert 203.0.113.10

This means the first router responded but the next hop did not respond to the probes. It does not necessarily prove that the router is down; the next router may filter or not respond to traceroute probes.

Step 3: Troubleshooting sequence

Check the PC

ipconfig

ping 192.168.10.1

Verify the IP address, subnet mask and gateway.

Check the router

show ip interface brief

show ip route

Confirm the interfaces are up and a route to the server exists.

Check the destination

- Verify the server IP and subnet mask.
- Verify the server's default gateway.
- Confirm DNS records and the HTTP service are configured.

Step 4: Example resolution

Suppose R1 has no route to the remote server network. For a simple lab with one upstream router at 198.51.100.2, configure:

enable

configure terminal

ip route 203.0.113.0 255.255.255.0 198.51.100.2

end

show ip route

Only use this command if that next hop is actually connected and the remote server network is reachable through it. In a different topology, configure the appropriate route instead.

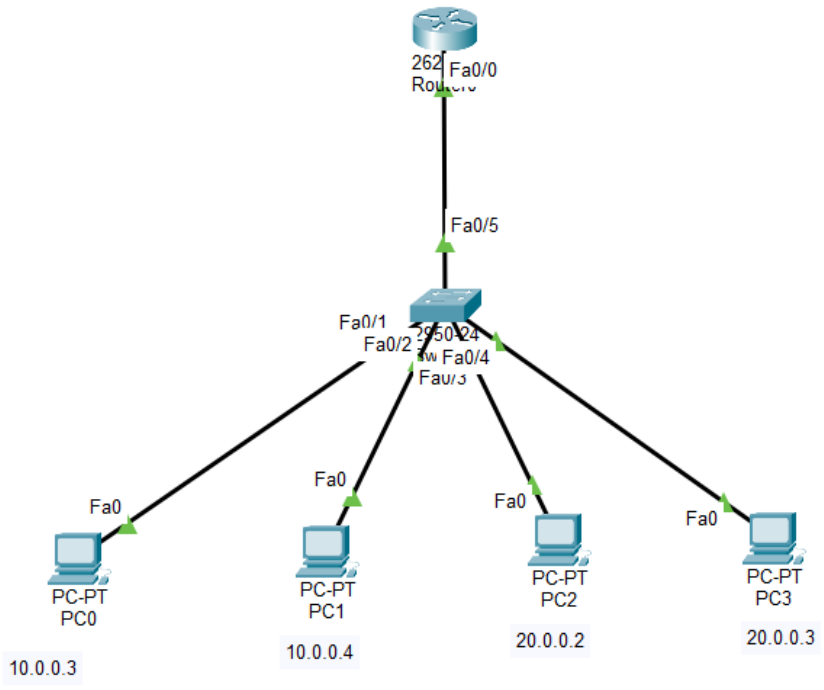
Finally, repeat tracert, ping the server, and test the web browser.

Lab report conclusion: Traceroute helped narrow down the point where the path stopped responding. The route, DNS configuration and server settings were checked, and the identified misconfiguration was corrected. Record the actual failing hop from your own lab rather than claiming that a particular hop failed without evidence.

Task 4.

Simulate a scenario where a router has an incorrect static route that breaks connectivity between two VLANs. Correct the route configuration and verify the solution using appropriate show commands.

Hint: Use 'show ip route' and 'show running-config' to check current routes and settings.



```
Router#
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    10.0.0.0/8 is directly connected, FastEthernet0/0.10
C    20.0.0.0/8 is directly connected, FastEthernet0/0.20
```

```
!
!
interface FastEthernet0/0
 no ip address
 duplex auto
 speed auto
!
interface FastEthernet0/0.10
 encapsulation dot1Q 10
 ip address 10.0.0.1 255.0.0.0
!
interface FastEthernet0/0.20
 encapsulation dot1Q 20
 ip address 20.0.0.1 255.0.0.0
!
interface FastEthernet0/1
 no ip address
 duplex auto
 speed auto
!
interface Serial0/0
 no ip address
 clock rate 2000000
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
 login
!
!
!
end
```

5. Troubleshoot WhatsApp voice calls between two branches

Objective

Analyze a multi-router network where WhatsApp voice calls fail between two branches, identify a possible misconfiguration, and document the troubleshooting and resolution.

Since you haven't supplied the network diagram or running configurations, the following is a sample scenario, not a confirmed diagnosis of your assignment's specific fault.

Step 1: Check the router configurations

On each router, collect:

show ip interface brief

show ip route

show running-config

show access-lists

Look for:

- Incorrect IP addresses or subnet masks on WAN interfaces.
- Missing routes to the other branch's LAN.
- Missing return routes.
- ACLs denying required traffic.
- NAT or firewall policies interfering with voice traffic.

Step 2: Test basic connectivity first

From a PC at Branch A:

ping 192.168.10.1

ping 192.168.20.1

tracert 192.168.20.10

Use the actual gateway and destination addresses from your topology.

If the remote gateway or PC is unreachable, investigate routing and filtering before diagnosing the voice application. A successful ping alone does not prove that voice traffic is permitted.

Step 3: Example misconfiguration — ACL blocking traffic

Suppose R2 contains:

access-list 101 deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255

access-list 101 permit ip any any

The first rule denies all IP traffic from Branch A's subnet to Branch B's subnet. This would block much more than voice calls.

If this is the demonstrated fault in your lab, and the intended policy is to permit inter-branch traffic, remove the incorrect entry and verify the ACL again.

configure terminal

no access-list 101 deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255

end

show access-lists

Important: This is an example only. Do not remove a production security rule without authorization. In a real network, replace an incorrect rule with a policy that permits only the required traffic.

Step 4: Verify routing and voice traffic

WhatsApp calls rely on Internet connectivity and encrypted, dynamically negotiated voice/media traffic. They do not necessarily use one fixed UDP port in every call. Therefore, don't assume that a single port is always the cause.

Check:

1. Whether both branches have working Internet access.
2. Whether DNS resolution works.
3. Whether NAT is configured correctly at the Internet edge.
4. Whether firewall rules permit the application's required traffic.
5. Whether routing and return traffic work in both directions.
6. Whether the issue affects all Internet access or only voice calls.

Packet Tracer can simulate IP routing, ACLs and basic services, but it cannot fully reproduce a real WhatsApp voice call. Use a real, authorized test environment or suitable packet captures to investigate application-specific media failures.

Step 5: Record the outcome

After making the verified correction, repeat the ping and traceroute tests, check ACL counters, and test the application again in the actual environment.

Lab report conclusion: The router configurations were inspected, connectivity and routing were tested, and ACL, NAT and return-path behavior were investigated. The specific fault and resolution must be based on the supplied configurations and observed test results.